5. OpenCV pixel operation

Pixel operation, we can change the pixel color at any position to a new one. Here we first read the image and then assign an area to white.

Code path:

```
~/dofbot_ws/src/dofbot_opencv/scripts/1.OpenCV_basic/04_pixel_pic.ipynb
```

The main code is as follows:

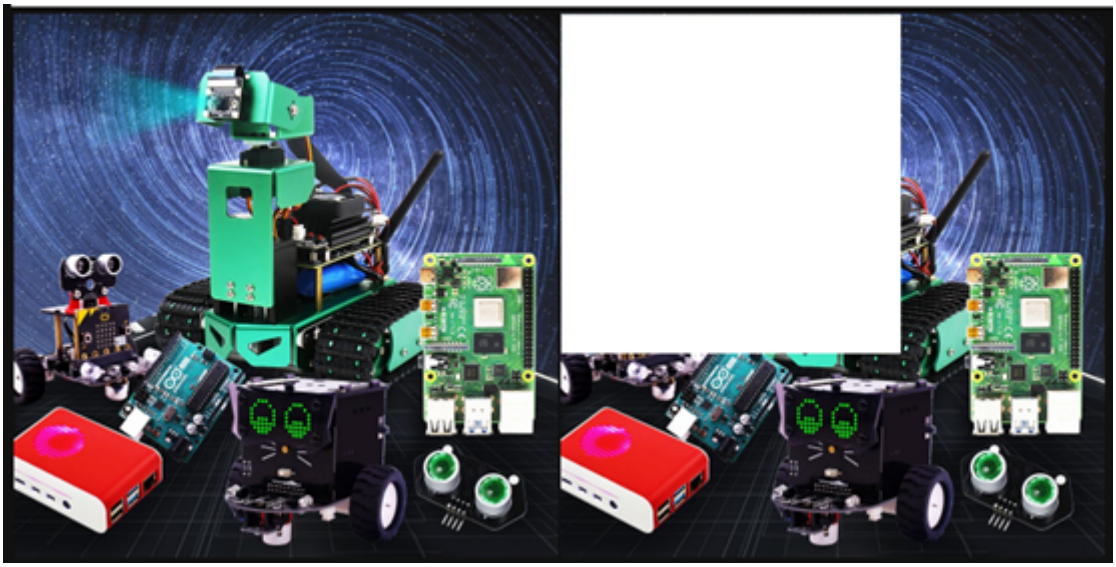
```
import cv2

img = cv2.imread('yahboom.jpg',1)
(b,g,r) = img[100,100]
print(b,g,r)# bgr
#10 100 --- 110 100
i=j=0
for j in range(1,500):
      img[i,j] = (255,255,255)
      for i in range(1,500):
        img[i,j] = (255,255,255)
# cv2.imshow('image',img)
# cv2.waitKey(0) #1000 ms
```

```
#bgr8转jpeg格式
import enum
import cv2
def bgr8_to_jpeg(value, quality=75):
      return bytes(cv2.imencode('.jpg', value)[1])
```

JupyterLab shows the image comparison before and after:

```
import ipywidgets.widgets as widgets
image_widget1 =   widgets.Image(format='jpg', )
image_widget2 =   widgets.Image(format='jpg', )
# create a horizontal box container   to place the image widget next to
eachother
image_container =   widgets.HBox([image_widget1, image_widget2])
# display the container in this   cell's output
display(image_container)
img1 = cv2.imread('yahboom.jpg',1)
image_widget1.value =   bgr8_to_jpeg(img1)  #original

image_widget2.value =   bgr8_to_jpeg(img)   #After pixel operation
```

After the code block is run, you can see that part of the second image has turned into white pixels.